```
// ==========================================
// INSTAGRAM-LIKE DATABASE USING MONGODB
// ==========================================

// 1. Create / Select Database
use instagram_db


// ==========================================
// 2. USERS COLLECTION
// ==========================================

db.users.insertMany([
  {
    user_id: 1,
    username: "simran",
    name: "Simran Kaur",
    email: "simran@gmail.com",
    followers: 1200,
    following: 350
  },
  {
    user_id: 2,
    username: "aman",
    name: "Aman Singh",
    email: "aman@gmail.com",
    followers: 850,
    following: 420
  },
  {
    user_id: 3,
    username: "priya",
    name: "Priya Sharma",
    email: "priya@gmail.com",
    followers: 2100,
    following: 500
  },
  {
    user_id: 4,
    username: "rahul",
    name: "Rahul Kumar",
    email: "rahul@gmail.com",
    followers: 650,
    following: 300
  },
```

```
  {
    user_id: 5,
    username: "neha",
    name: "Neha Verma",
    email: "neha@gmail.com",
    followers: 1800,
    following: 450
  }
])


// ============================================
// 3. POSTS COLLECTION
// ============================================

db.posts.insertMany([
  {
    post_id: 101,
    username: "simran",
    caption: "Beautiful day!",
    hashtag: "#nature",
    likes: 250,
    comments: 30,
    location: "Chandigarh"
  },
  {
    post_id: 102,
    username: "aman",
    caption: "Learning MongoDB today",
    hashtag: "#mongodb",
    likes: 180,
    comments: 20,
    location: "Delhi"
  },
  {
    post_id: 103,
    username: "priya",
    caption: "Amazing travel experience",
    hashtag: "#travel",
    likes: 520,
    comments: 75,
    location: "Manali"
  },
  {
    post_id: 104,
```

```
    username: "rahul",
    caption: "Big Data practical",
    hashtag: "#bigdata",
    likes: 150,
    comments: 15,
    location: "Amritsar"
  },
  {
    post_id: 105,
    username: "neha",
    caption: "Beautiful sunset",
    hashtag: "#nature",
    likes: 430,
    comments: 50,
    location: "Goa"
  },
  {
    post_id: 106,
    username: "simran",
    caption: "MongoDB practical work",
    hashtag: "#mongodb",
    likes: 320,
    comments: 40,
    location: "Chandigarh"
  }
])


// ============================================
// 4. COMMENTS COLLECTION
// ============================================

db.comments.insertMany([
  {
    comment_id: 1,
    post_id: 101,
    username: "aman",
    comment: "Beautiful picture!"
  },
  {
    comment_id: 2,
    post_id: 101,
    username: "priya",
    comment: "Amazing!"
  },
```

```
    {
        comment_id: 3,
        post_id: 103,
        username: "simran",
        comment: "Nice place!"
    },
    {
        comment_id: 4,
        post_id: 105,
        username: "rahul",
        comment: "Wonderful sunset!"
    }
])


// ============================================
// 5. DISPLAY ALL USERS
// ============================================

db.users.find()


// ============================================
// 6. DISPLAY ALL POSTS
// ============================================

db.posts.find()


// ============================================
// 7. FIND POSTS BY HASHTAG
// ============================================

db.posts.find({
    hashtag: "#nature"
})


// ============================================
// 8. FIND POSTS WITH MORE THAN 300 LIKES
// ============================================

db.posts.find({
    likes: {$gt: 300}
})
```

```
// ==========================================
// 9. FIND POSTS BY SIMRAN
// ==========================================

db.posts.find({
   username: "simran"
})


// ==========================================
// 10. SORT POSTS BY LIKES
// ==========================================

db.posts.find().sort({
   likes: -1
})


// ==========================================
// 11. TOP 3 MOST LIKED POSTS
// ==========================================

db.posts.find()
   .sort({likes: -1})
   .limit(3)


// ==========================================
// 12. FIND USERS WITH MORE THAN 1000 FOLLOWERS
// ==========================================

db.users.find({
   followers: {$gt: 1000}
})


// ==========================================
// 13. UPDATE LIKES
// ==========================================

db.posts.updateOne(
   {post_id: 101},
   {$set: {likes: 300}}
```

```
)


// ============================================
// 14. INCREASE FOLLOWERS
// ============================================

db.users.updateOne(
   {username: "simran"},
   {$inc: {followers: 100}}
)


// ============================================
// 15. CREATE INDEX ON HASHTAG
// ============================================

db.posts.createIndex({
   hashtag: 1
})


// ============================================
// 16. CREATE INDEX ON USERNAME
// ============================================

db.posts.createIndex({
   username: 1
})


// ============================================
// 17. VIEW ALL INDEXES
// ============================================

db.posts.getIndexes()


// ============================================
// 18. SEARCH USING INDEXED HASHTAG
// ============================================

db.posts.find({
   hashtag: "#mongodb"
})
```

```
// ============================================
// 19. SEARCH USING INDEXED USERNAME
// ============================================

db.posts.find({
    username: "simran"
})


// ============================================
// 20. REMOVE HASHTAG INDEX
// ============================================

db.posts.dropIndex("hashtag_1")


// ============================================
// 21. VIEW INDEXES AGAIN
// ============================================

db.posts.getIndexes()
```